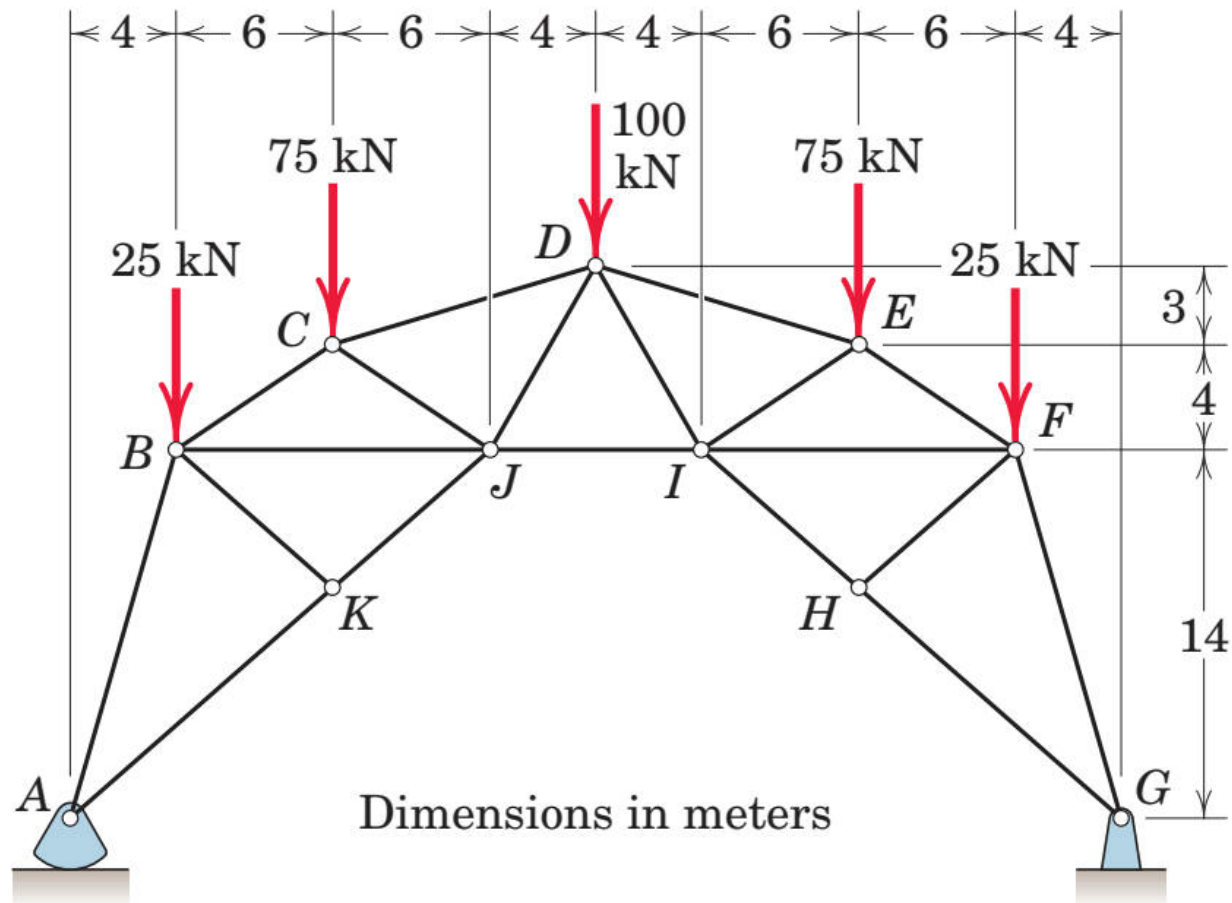


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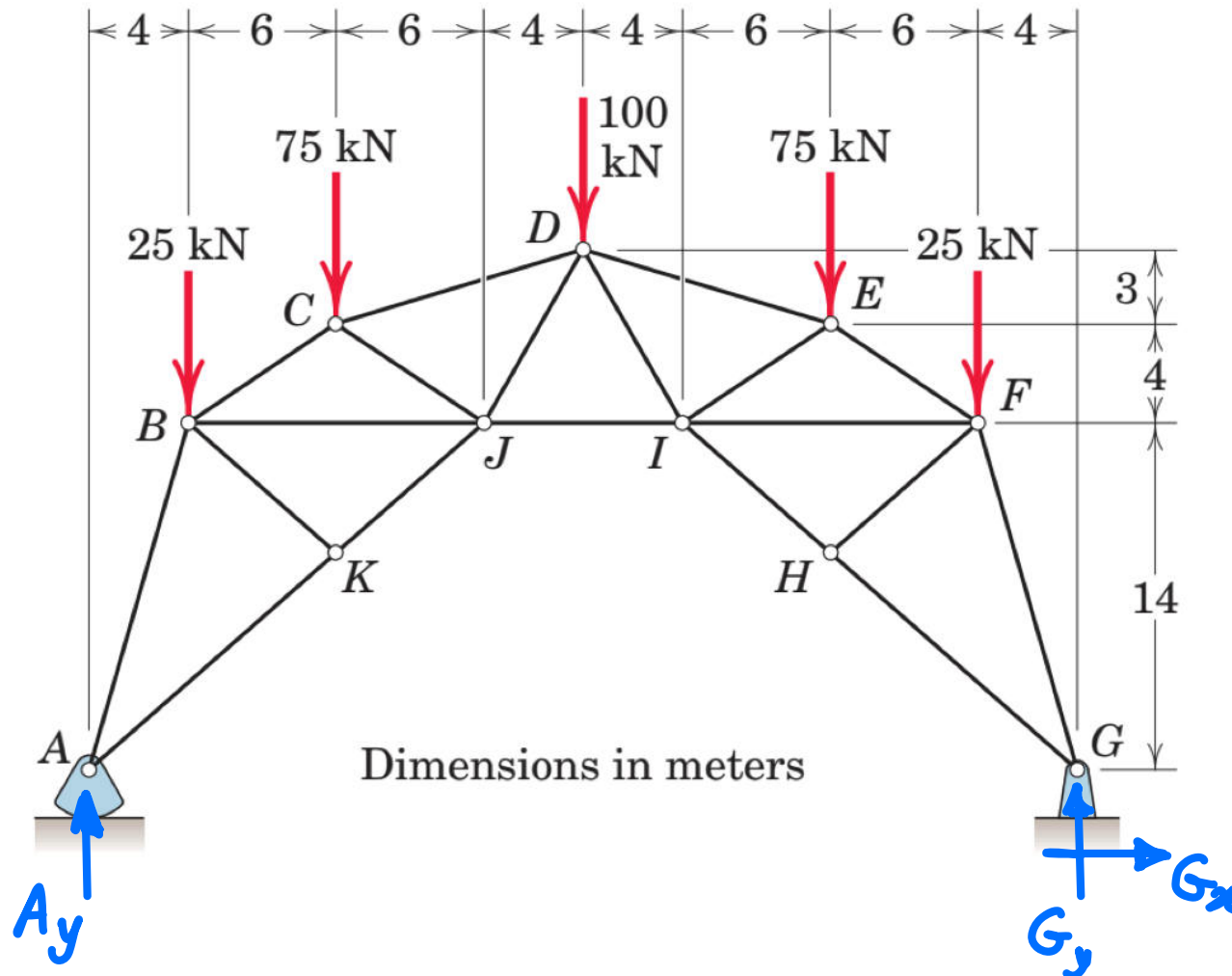
Dep. of Mechanical Engineering

Statics (ME 231)
Review Problems
By Saeed Lotfan

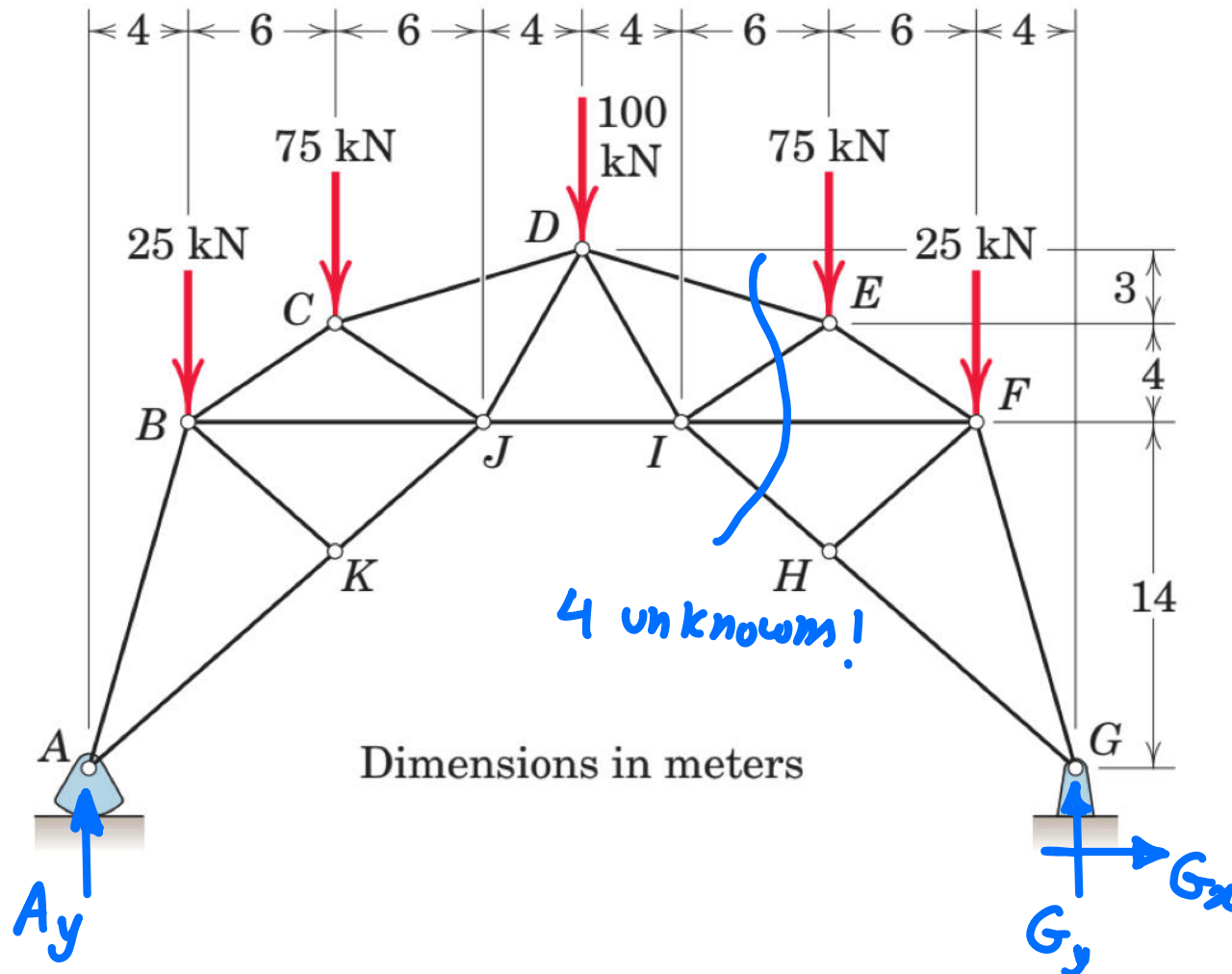
P1. Determine the forces in members DE , EI , FI , and HI of the arched roof truss below.



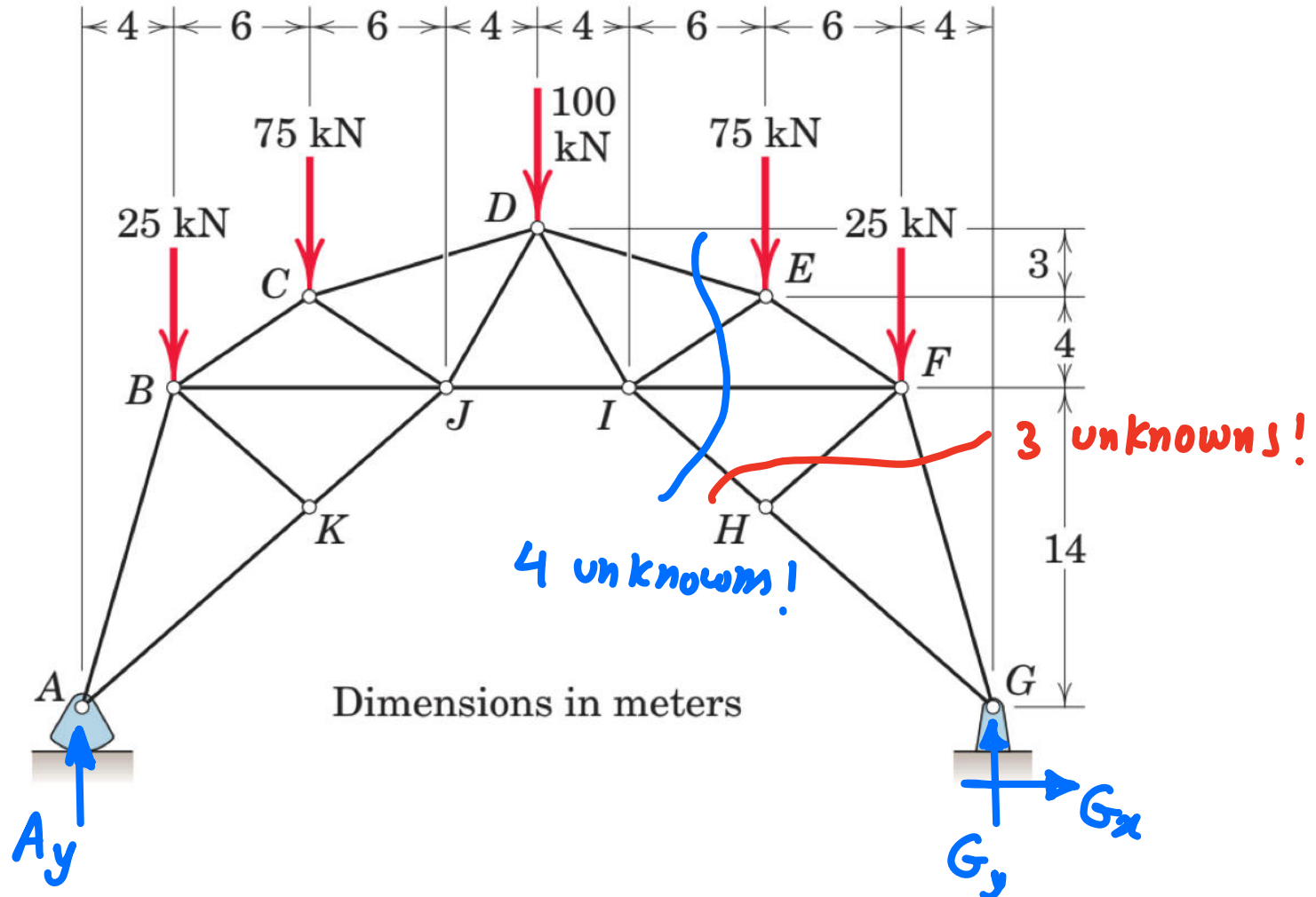
Solution P1. $A_y = G_y = 150 \text{ kN}$, $G_x = 0$



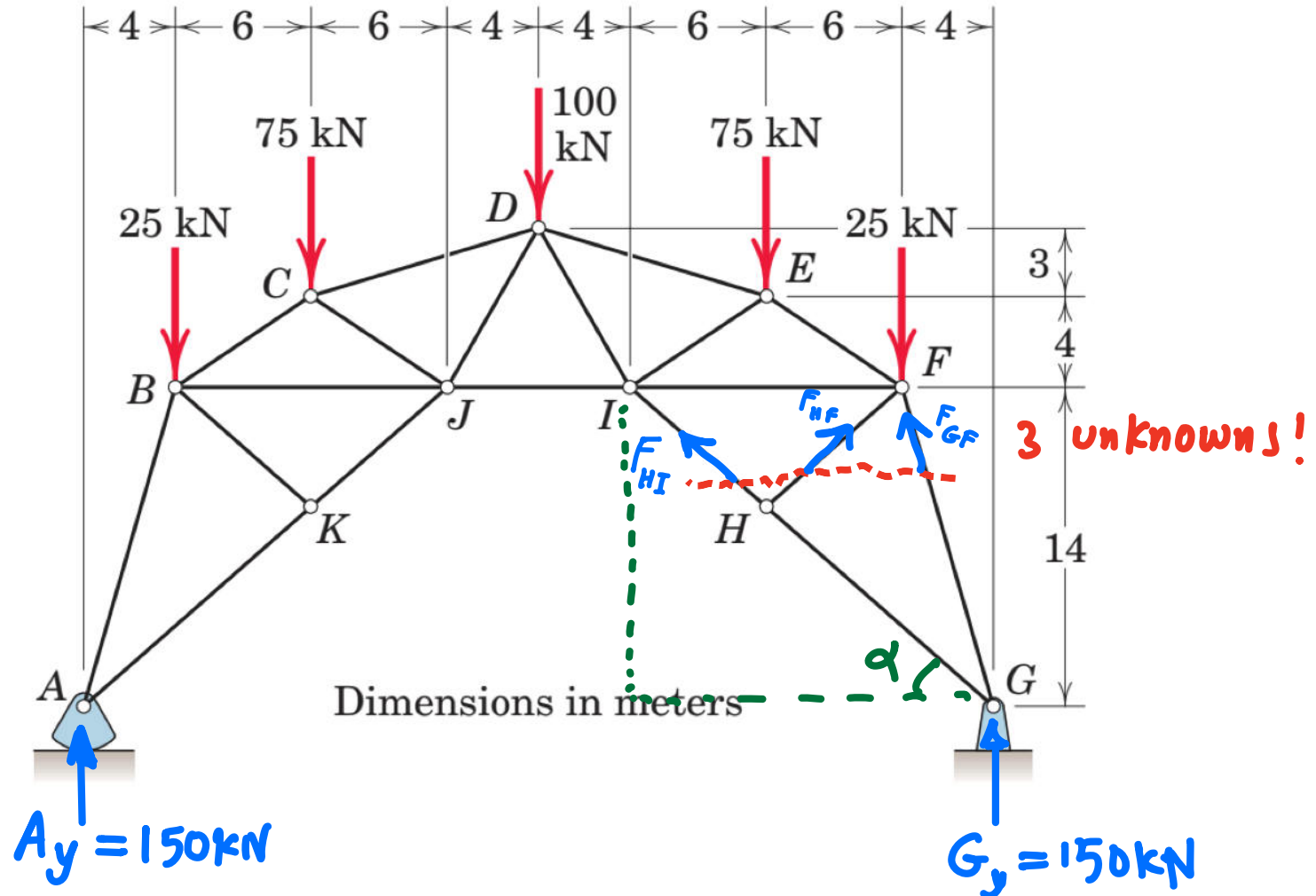
Solution P1. $A_y = G_y = 150 \text{ kN}$, $G_x = 0$



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Solution P1. $\sum M_F = 0$



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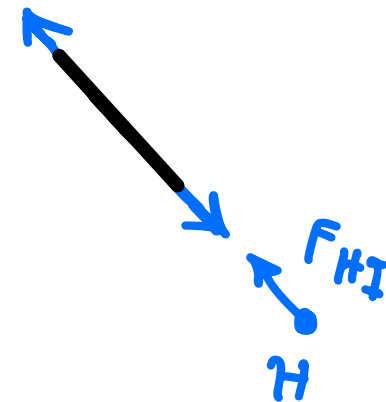
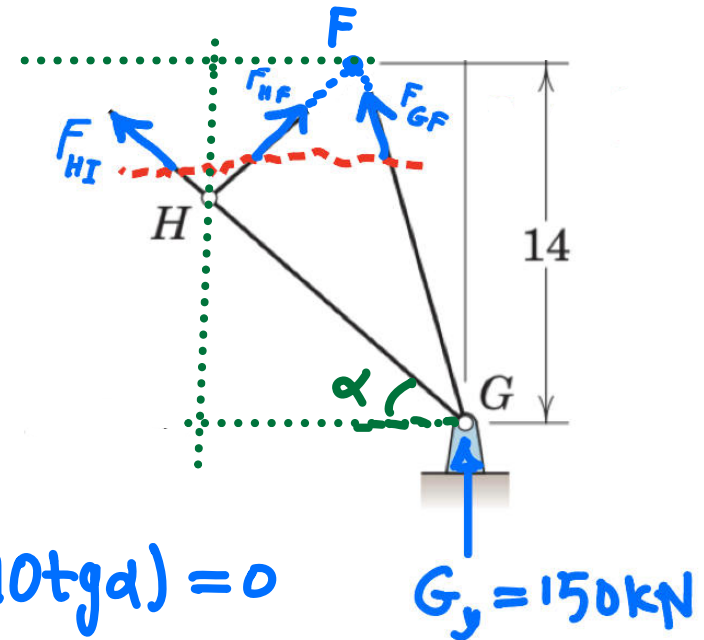
$$\alpha = \tan^{-1}\left(\frac{14}{16}\right) = 41.19^\circ$$

$$\uparrow (\sum M_F = G_y(4) - F_{HI} \sin \alpha (6)$$

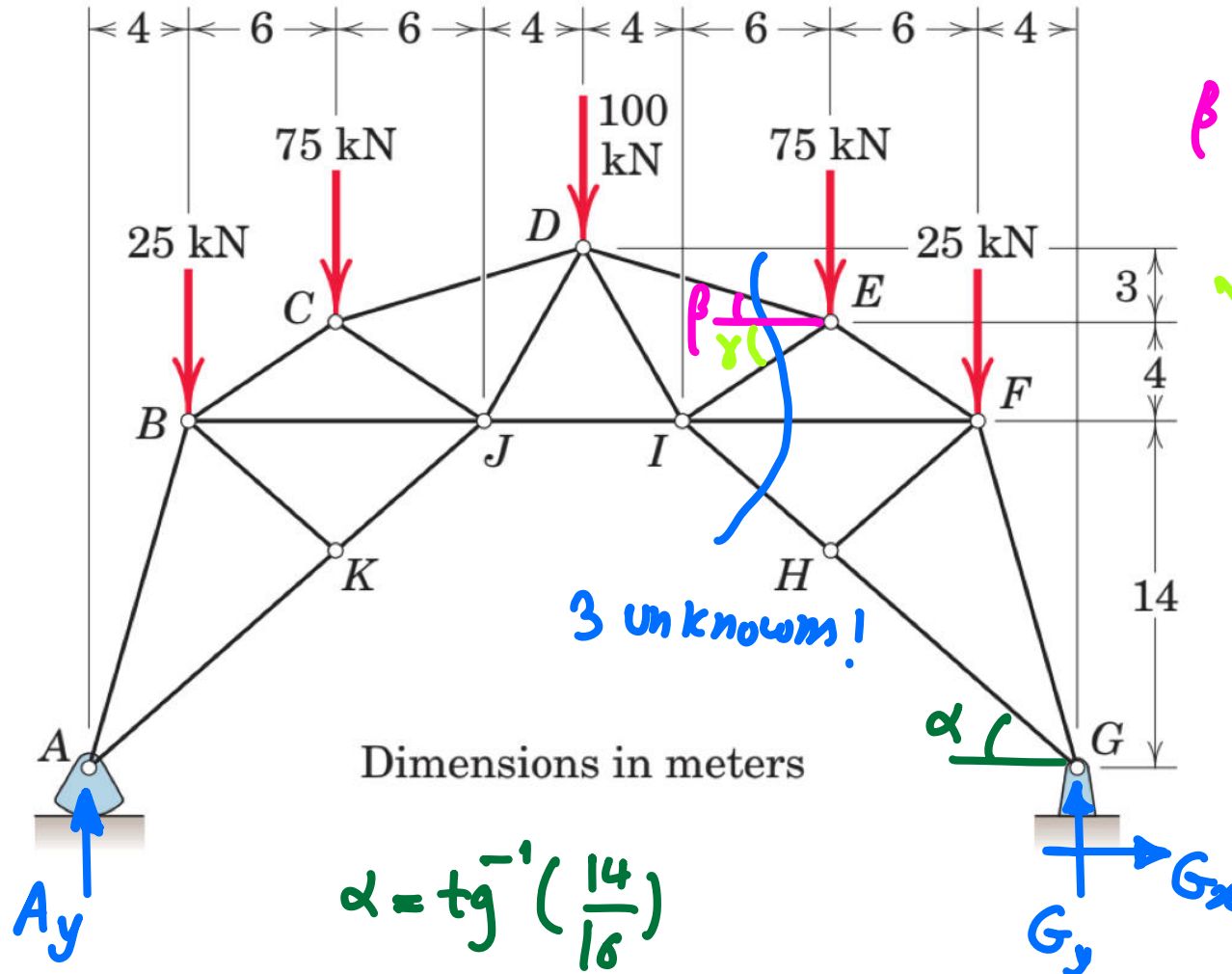
$$- F_{HI} \cos \alpha (14 - 10 \tan \alpha) = 0$$

$$\rightarrow F_{HI} = 75.94 \text{ kN}$$

HI is under Tension!



Solution P1. $A_y = G_y = 150 \text{ kN}$, $G_x = 0$



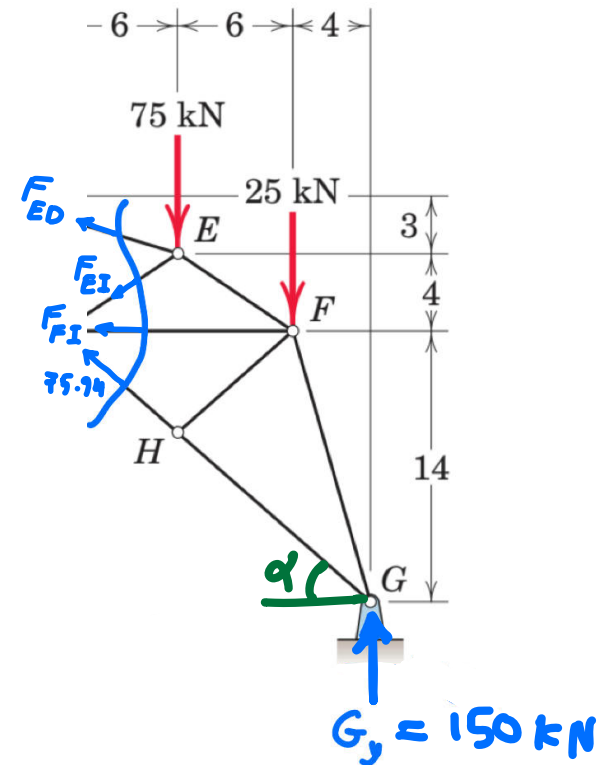
Solution P1.

$$+\circlearrowleft (\sum M_E = -F_{HI} \cos \alpha (4 + 14 - 10 + 9 \alpha)$$

$$-F_{FI} (4) - 25(6) + 150(10) = 0$$

$$\rightarrow F_{FI} = 205.36 \text{ kN}$$

FI is under Tension!



Solution P1.

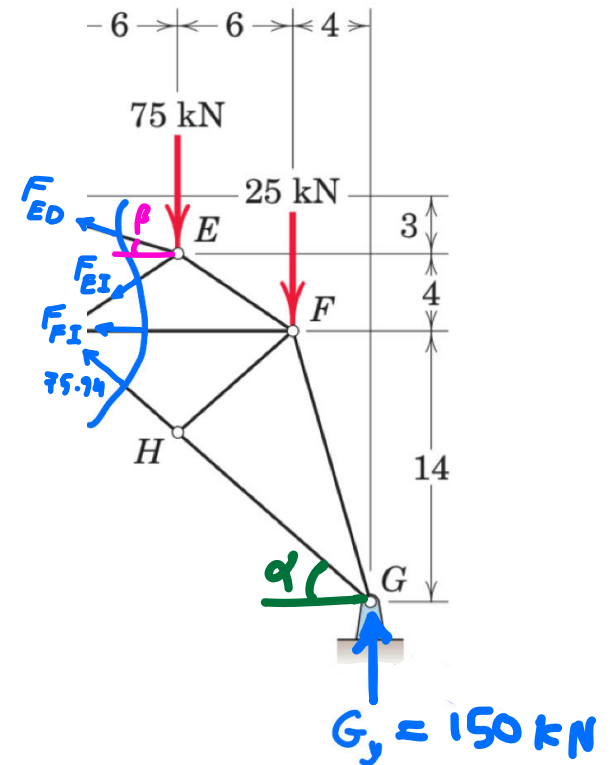
$$+\circlearrowleft \sum M_I = -75(6) - 25(12)$$

$$+ 150(16) + F_{ED} \cos \beta (4)$$

$$+ F_{ED} \sin \beta (8) = 0$$

$$F_{ED} = -297 \text{ kN}$$

ED is under Compression!



Solution P1.

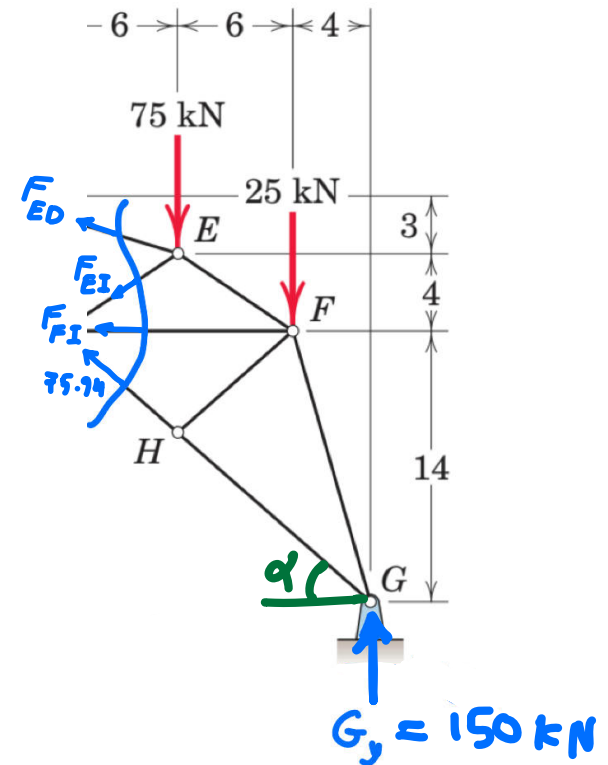
$$\sum F_x = 0 \quad \rightarrow$$

$$-75.94 \cos \alpha - F_{FI} - F_{ED} \cos \beta$$

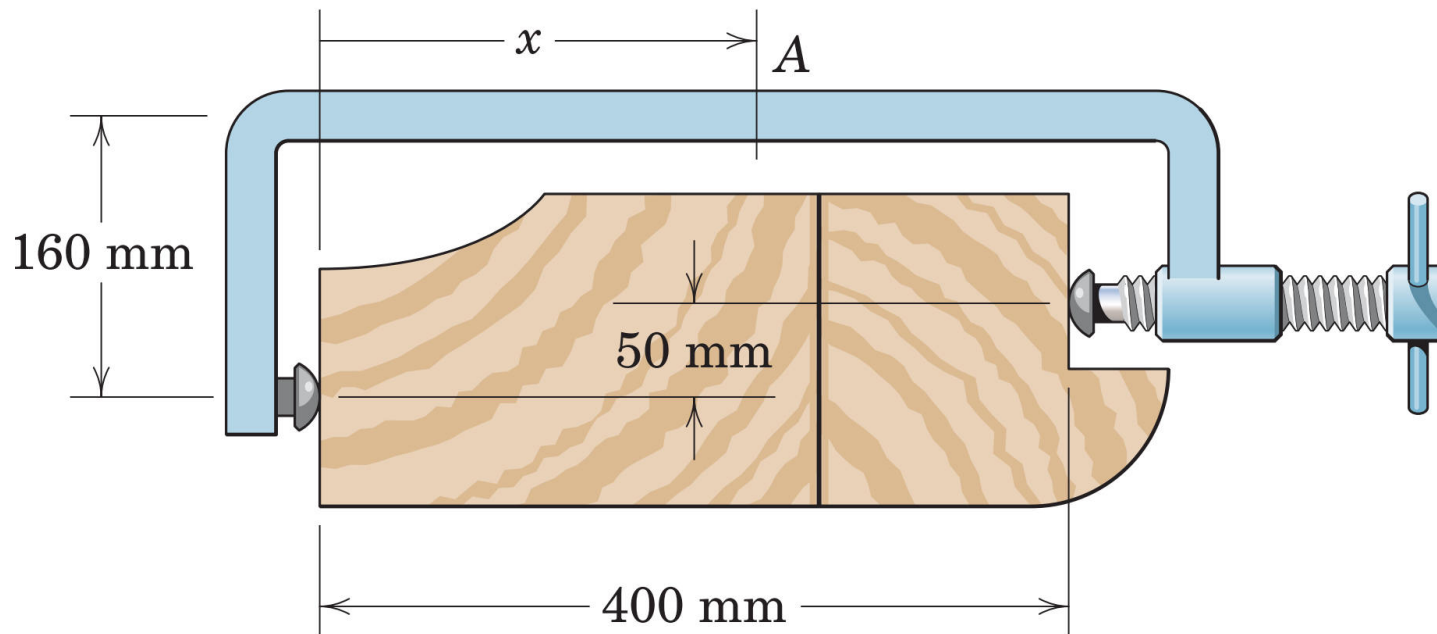
$$- F_{EI} \cos \gamma = 0$$

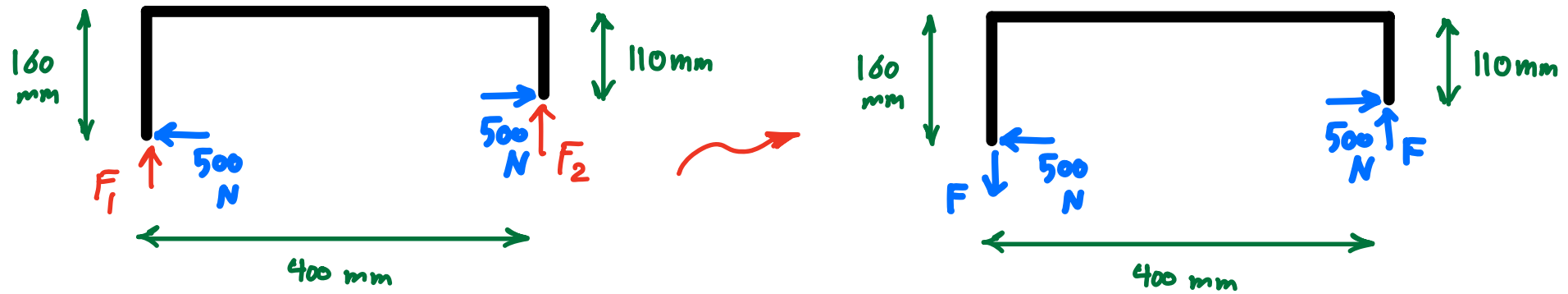
$$\rightarrow F_{EI} = 26.4 \text{ kN}$$

EI is under Tension!



P2. The adjusting screw of the special-purpose offset clamp supports a compression of 500 N. Calculate the shear force V , the tension T , and the bending moment M at section A of the clamp bar for $x = 250$ mm. Which of these three quantities changes with x ?

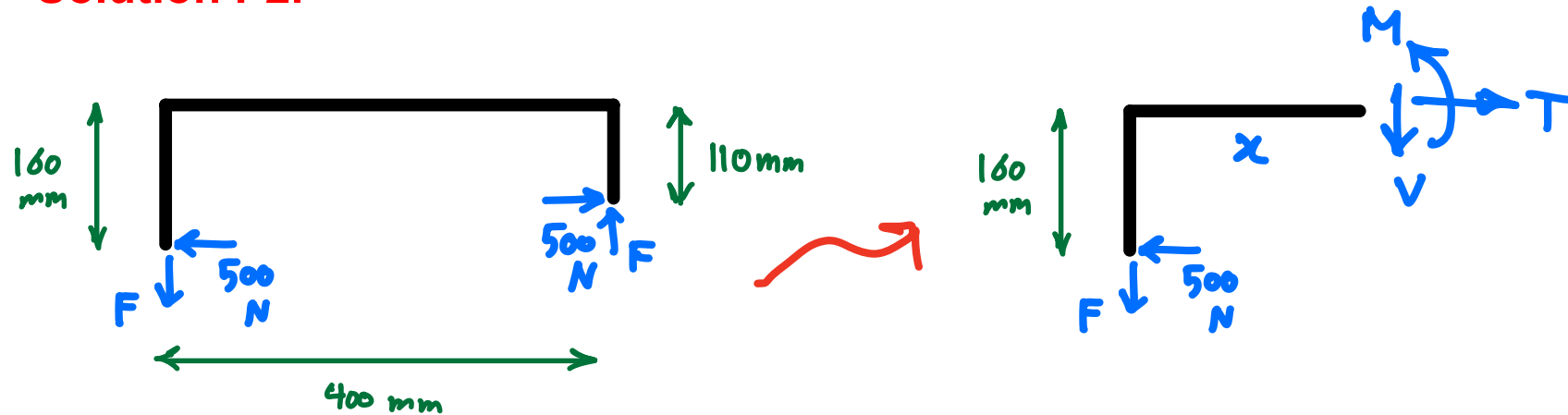


Solution P2.

$$F \times 400 = 500 \times 50$$

$$F = 62.5 \text{ N}$$

Solution P2.



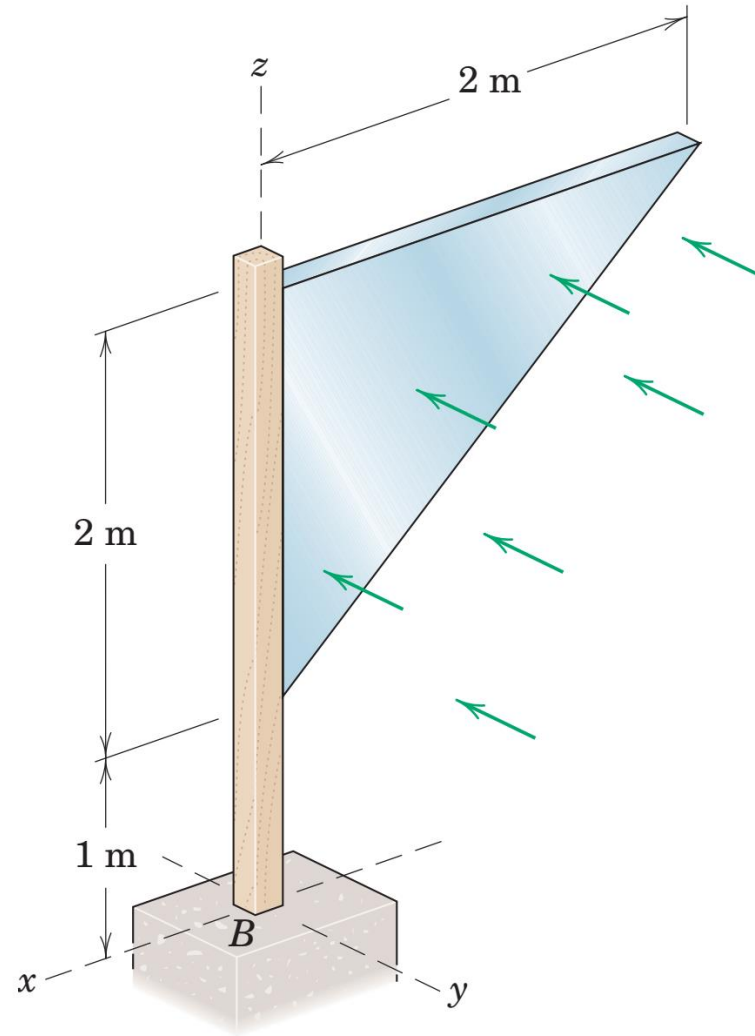
$$\sum F_x = 0 \rightarrow T = 500 \text{ N}$$

$$\sum F_y = 0 \rightarrow -V - F = 0 \rightarrow V = -F = -62.5 \text{ N}$$

$$\sum M_A = 0 \rightarrow M - T(160) - Vx = 0$$

$$M = (500 \times 160 - 62.5x) \text{ N}\cdot\text{mm}$$

P3. The triangular sign is attached to the post embedded in the concrete base at B . Calculate the shear force V , the bending moment M , and the torsional moment T in the post at B during a storm where the wind velocity normal to the sign reaches 100 km/h. The air pressure (called stagnation pressure) against the vertical surface corresponding to this velocity is 1.4 kPa.



Solution P3.

$$F_{wind} = PA = 1.4 \times \frac{1}{2}(2)(2) = 2.8 \text{ kN}$$

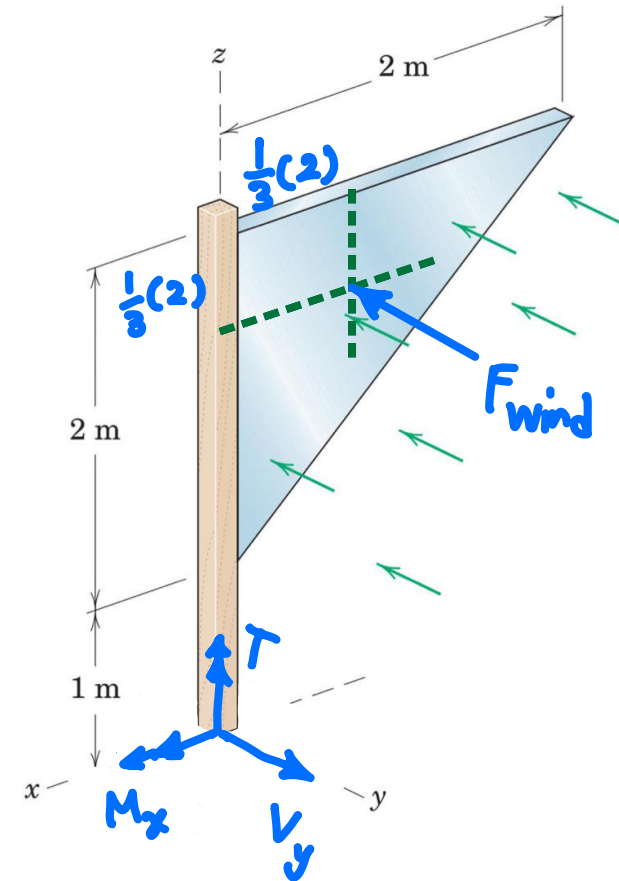
$$\sum F_y = 0 \rightarrow V_y = 2.8 \text{ kN}$$

$$\sum M_B^x = 0 \rightarrow F_w \left(\frac{4}{3} + 1 \right) + M_x = 0$$

$$M_x = -6.533 \text{ kN.m}$$

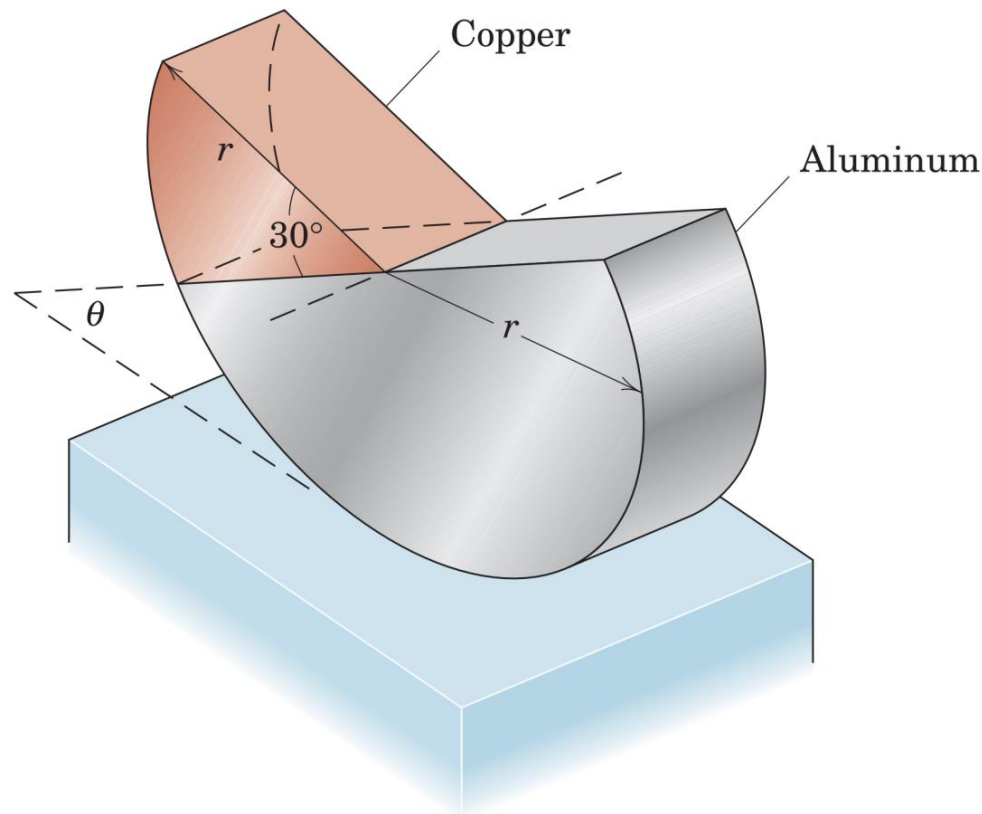
$$\sum M_B^z = 0 \rightarrow F_w \left(\frac{2}{3} \right) + T = 0$$

$$T = -1.866 \text{ kN.m}$$



what if we want to find internal loadings in the beam as $V(z)$, $M(z)$, $T(z)$?

P4. The 30° cylindrical sector is made of copper and is attached as shown to the semicylinder made of aluminum. Determine the angle θ for the equilibrium position of the cylinder resting on a horizontal surface.

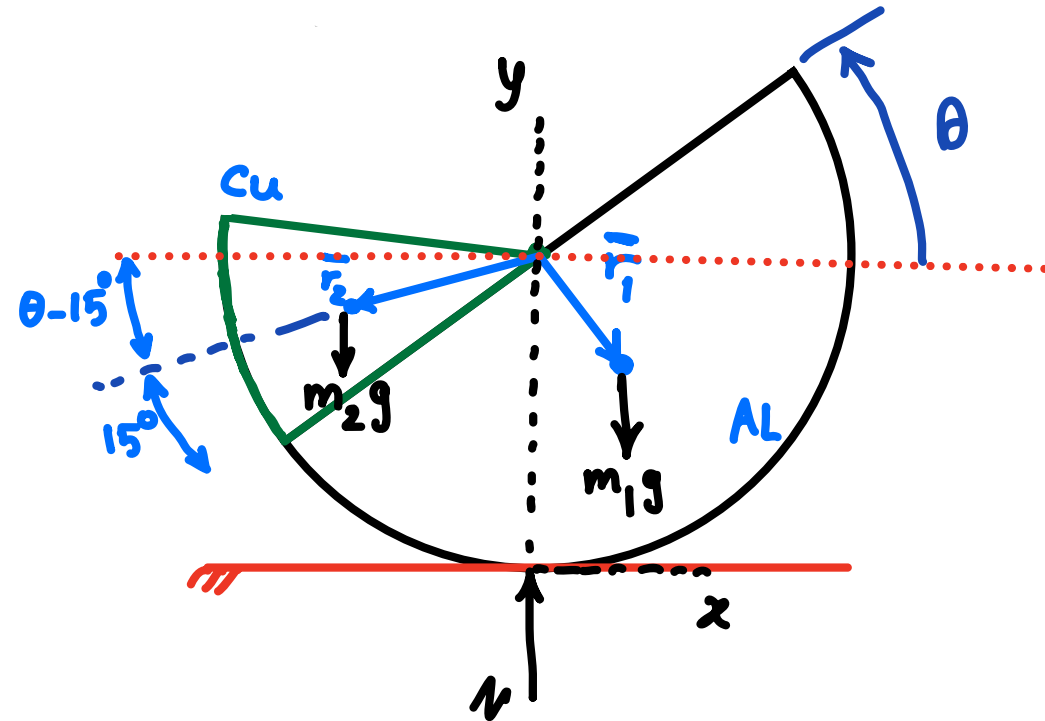


Solution P4.

$$\bar{r} = \frac{2r}{3} \frac{\sin \alpha}{\alpha}$$

$$\bar{r}_1 = \frac{2r}{3} \frac{\sin 90^\circ}{90^\circ \pi / 180^\circ} = \frac{4r}{3\pi}$$

$$\bar{r}_2 = \frac{2r}{3} \frac{\sin 15^\circ}{15^\circ \pi / 180} = \frac{8r}{\pi} \sin 15^\circ$$



Solution P4.

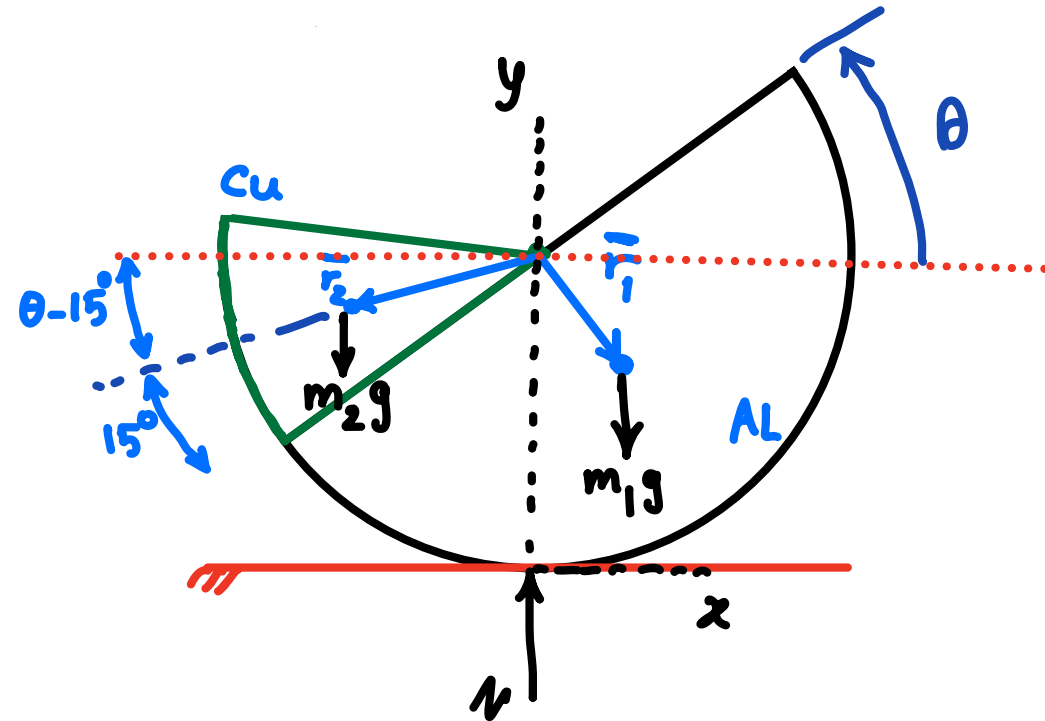
For equilibrium:

$$\bar{x} = 0$$

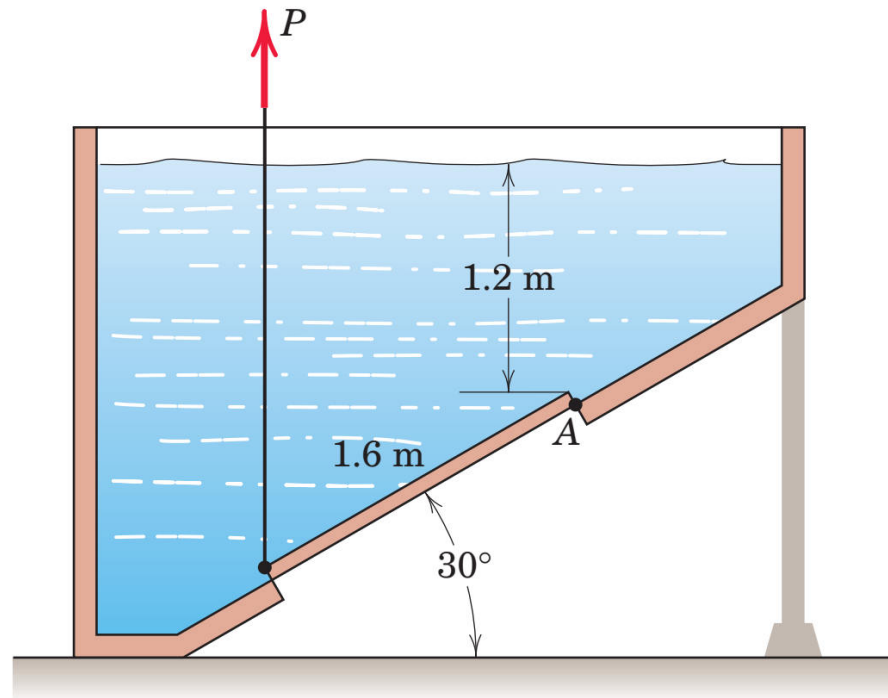
$$\bar{x} = \frac{\sum m_i \bar{x}_i}{\sum m_i} = 0$$

$$\bar{x} = \frac{m_1 \bar{x}_1 + m_2 \bar{x}_2}{m_{\text{tot}}}$$

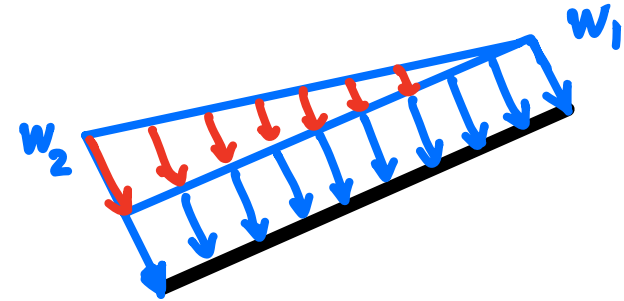
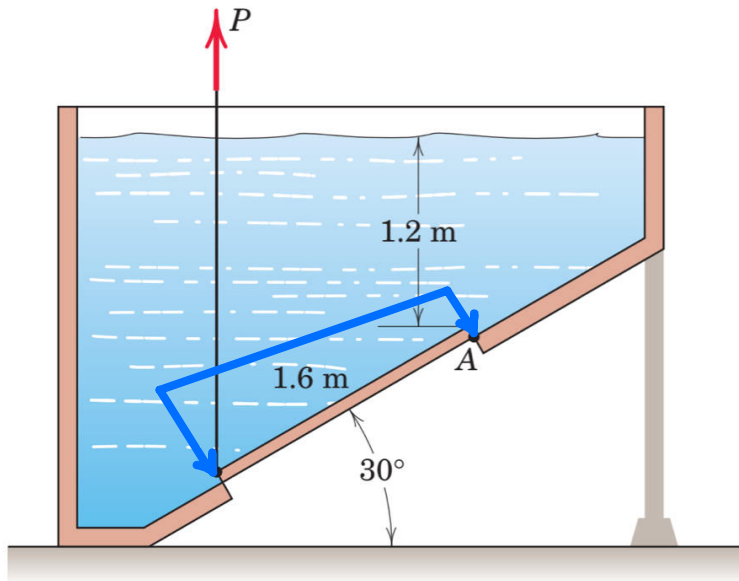
$$m_1 (\bar{r}_1 \cos(90^\circ - \theta)) - m_2 (\bar{r}_2 \cos(\theta - 15^\circ)) = 0$$



P5. The cross section of a fresh-water tank with a slanted bottom is shown. A rectangular door 1.6 m by 0.8 m (normal to the plane of the figure) in the bottom of the tank is hinged at A and is opened against the pressure of the water by the cable under a tension P as shown. Calculate P .



Solution P5.

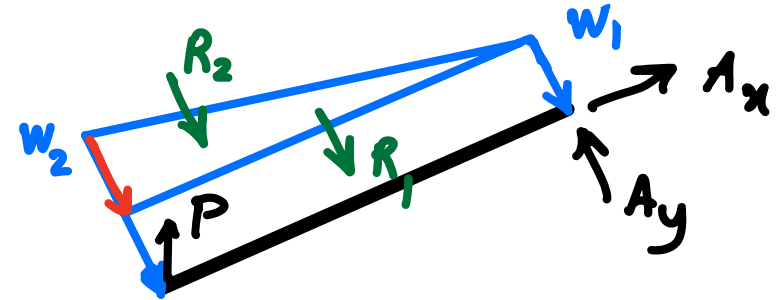
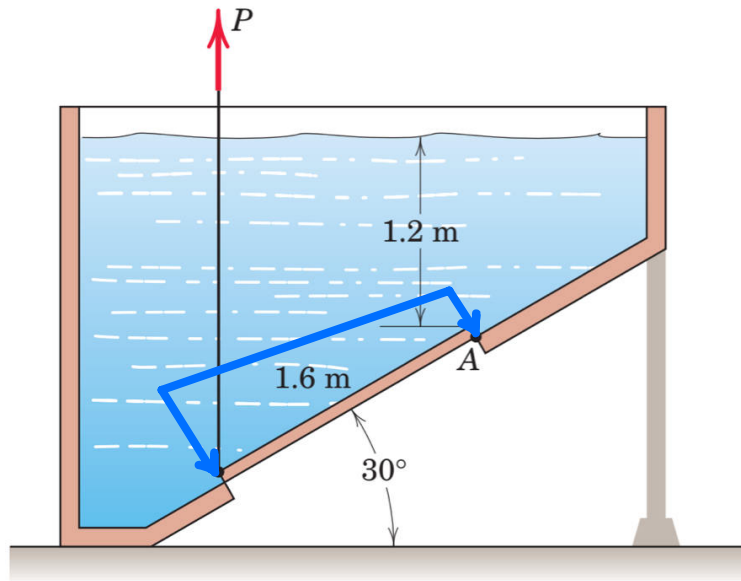


$$W = Pb = \rho_w g h b$$

$$W_1 = 1000 \times 10 \times 1.2 \times 0.8 = 9600 \text{ N/m}$$

$$W_2 = 1000 \times 10 \times (1.2 + 1.6 \sin 30) \times 0.8 = 16000 \text{ N/m}$$

Solution P5.



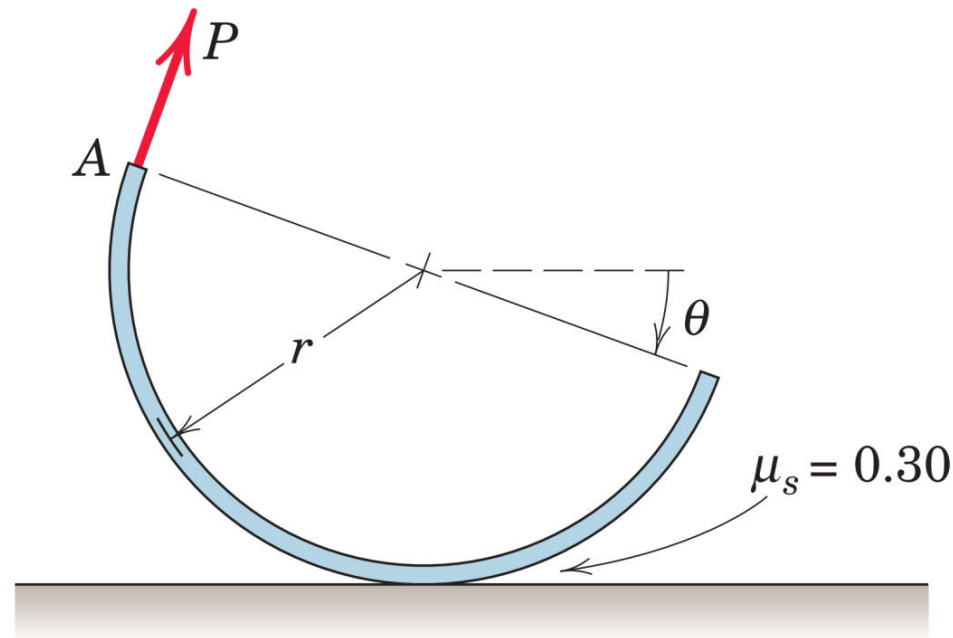
$$R_1 = W_1 (1.6) = 15360 \text{ N}$$

$$R_2 = (W_2 - W_1) \frac{1.6}{2} = 5120 \text{ N}$$

$$+\downarrow \sum M_A = 0 \rightarrow R_1 \left(\frac{1.6}{2} \right) + R_2 \left(\frac{2}{3} 1.6 \right) - P (1.6 \cos 30) = 0$$

$$P = 12809 \text{ N}$$

P6. The semicylindrical shell of mass m and radius r is rolled through an angle θ by the force P which remains tangent to its periphery at A as shown. If P is slowly increased, plot the tilt angle θ as a function of P up to the point of slipping. Determine the tilt angle $\theta_{\max}$ and the corresponding value $P_{\max}$ for which slipping occurs. The coefficient of static friction is 0.30



Solution P6.

$$+\rightarrow \sum F_x = 0$$

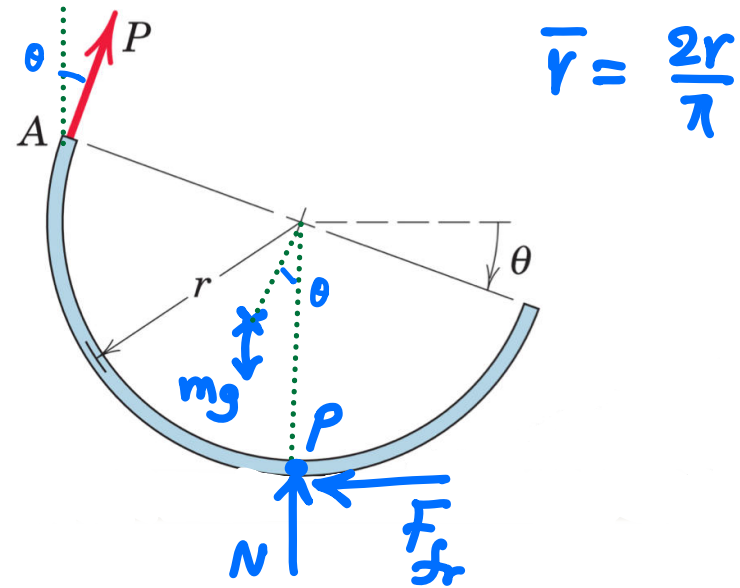
$$P \sin \theta - F_{fr} = 0$$

$$+\uparrow \sum F_y = 0$$

$$P \cos \theta - mg + N = 0$$

$$+\downarrow \sum M_p = 0$$

$$mg \bar{r} \sin \theta - P(r + r \sin \theta) = 0$$



Solution P6.

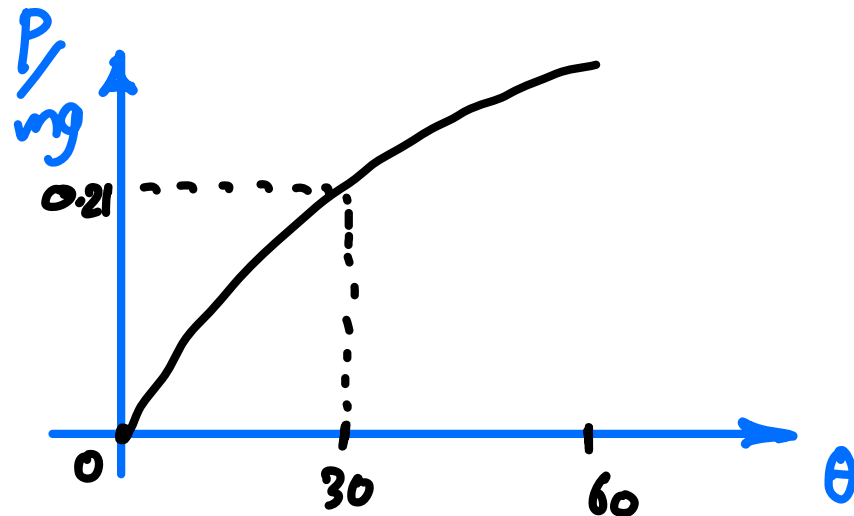
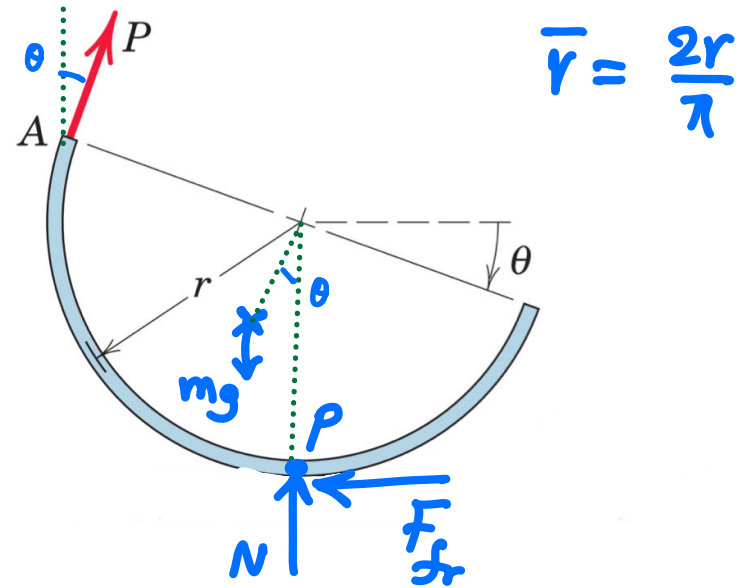
$$P \sin \theta - F_{fr} = 0$$

$$P \cos \theta - mg + N = 0$$

$$mg \bar{r} \sin \theta - P(r + r \sin \theta) = 0$$

$$P = \frac{2mg \sin \theta}{\pi(1 + \sin \theta)}$$

$$\frac{P}{mg} = \frac{2 \sin \theta}{\pi(1 + \sin \theta)}$$



Solution P6.

$$P \sin \theta - F_{fr} = 0$$

$$P \cos \theta - mg + N = 0$$

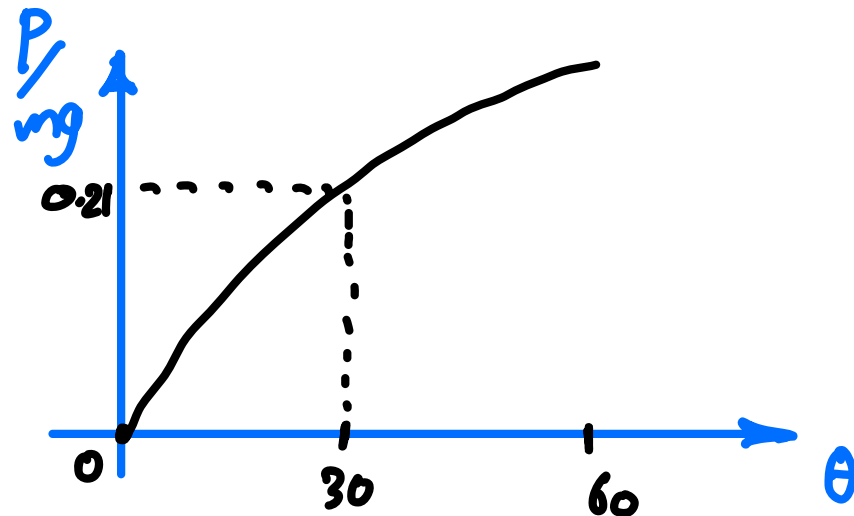
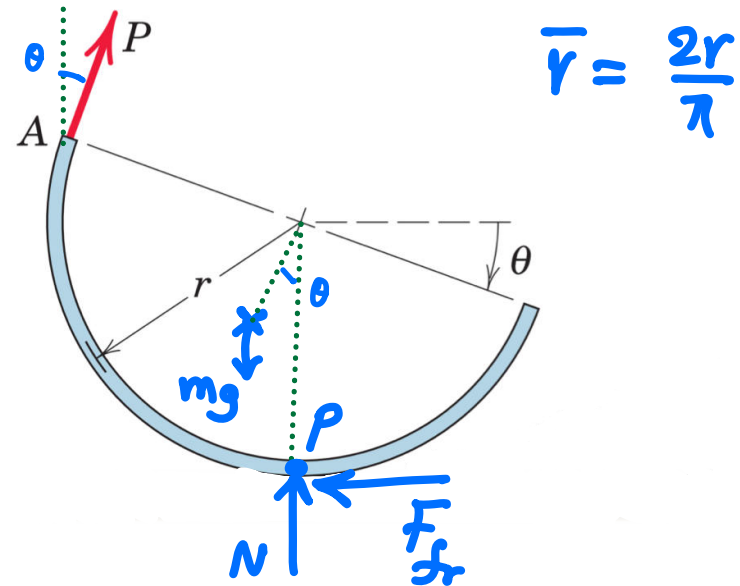
For θ_{max} ;

$$F_{fr} = \mu_s N$$

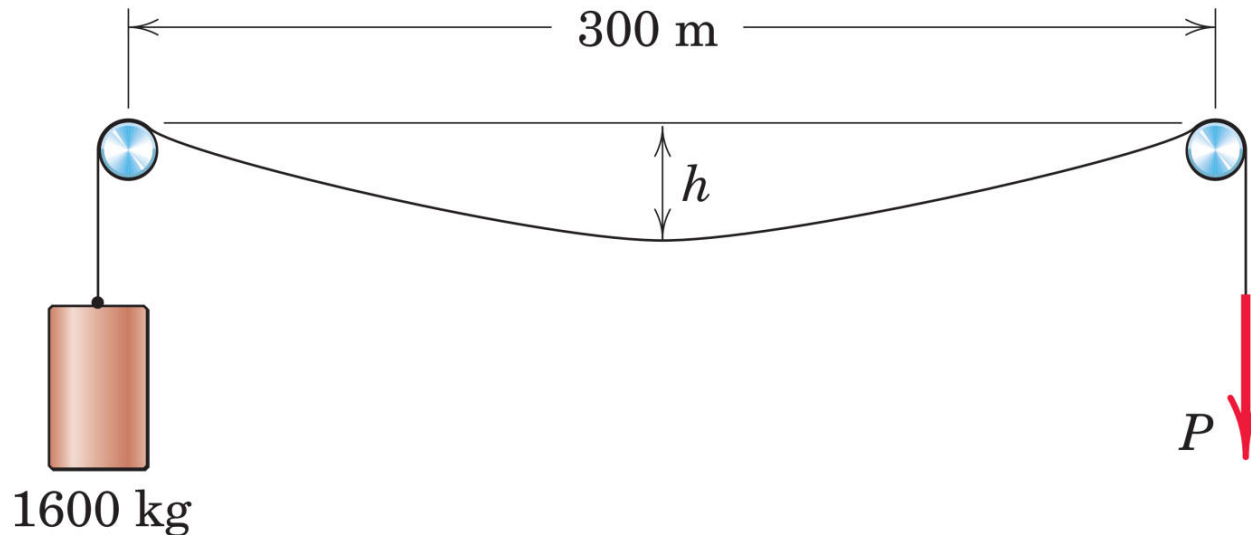
$$N = mg - P \cos \theta$$

$$N = mg - \frac{2 \sin \theta \cos \theta}{\pi (1 + \sin \theta)}$$

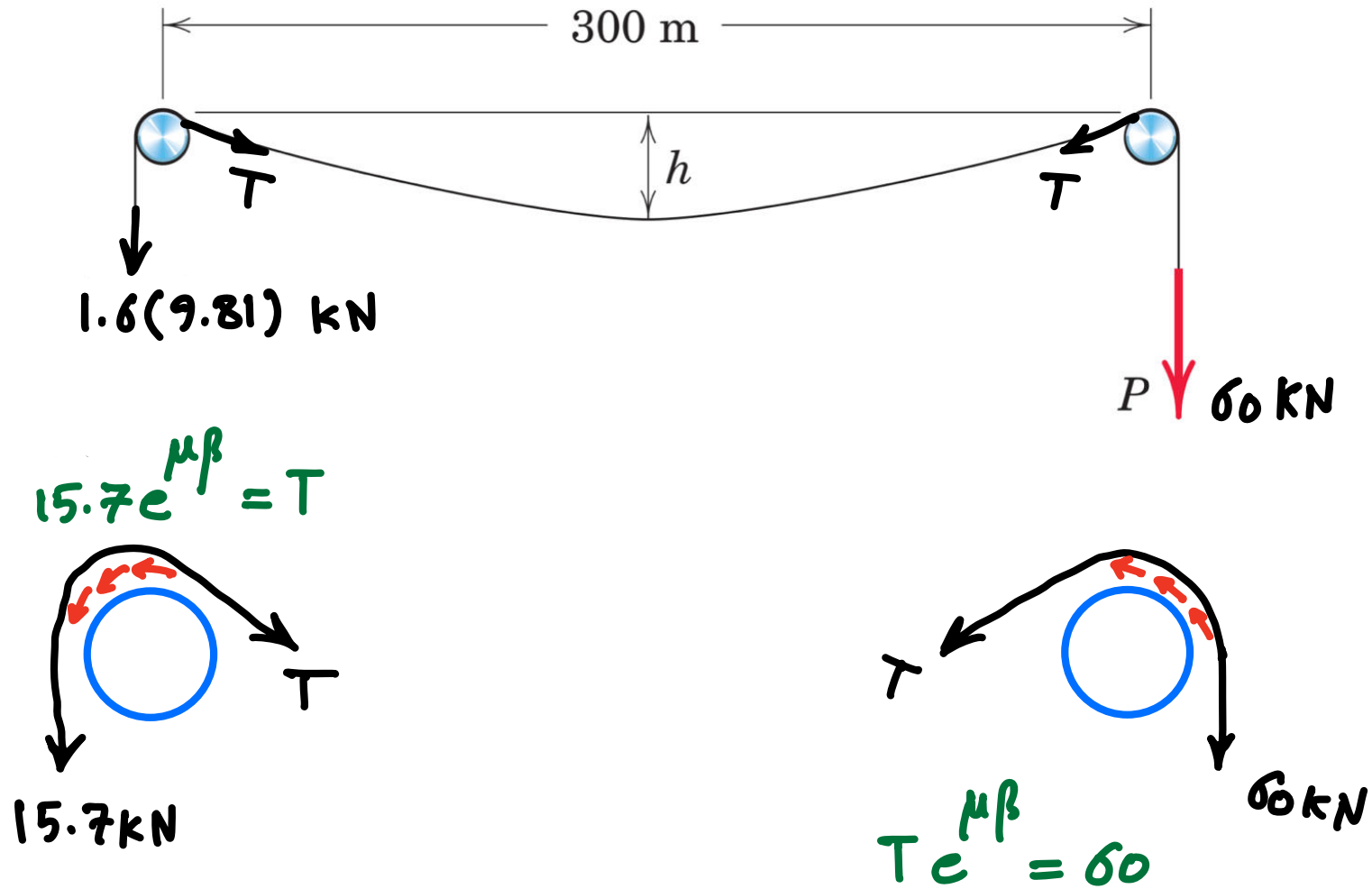
$$\theta_{max} = 59.9^\circ \quad P_{max} = 0.29 mg$$



P7. A heavy cable with a mass of 12 kg per meter of length passes over the two fixed pipes 300 m apart on the same level. One end supports a 1600-kg cylinder. By experiment it is found that a downward force P of 60 kN is required to induce slipping of the cable over both pipes at a constant rate. Determine the coefficient of kinetic friction between the cable and the pipes, the maximum tension T in the cable between the pipes, and the sag h in the cable.



Solution P7.

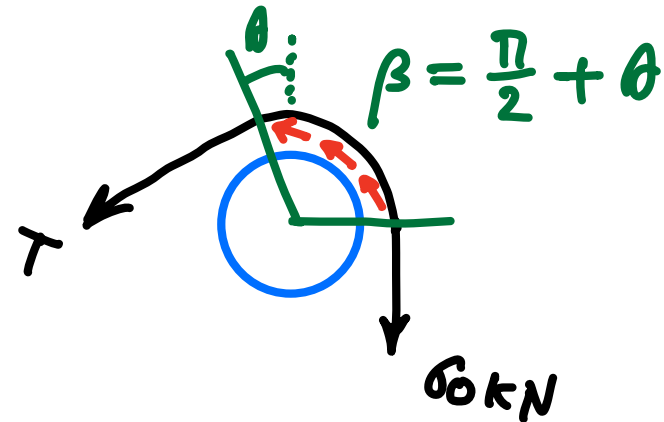


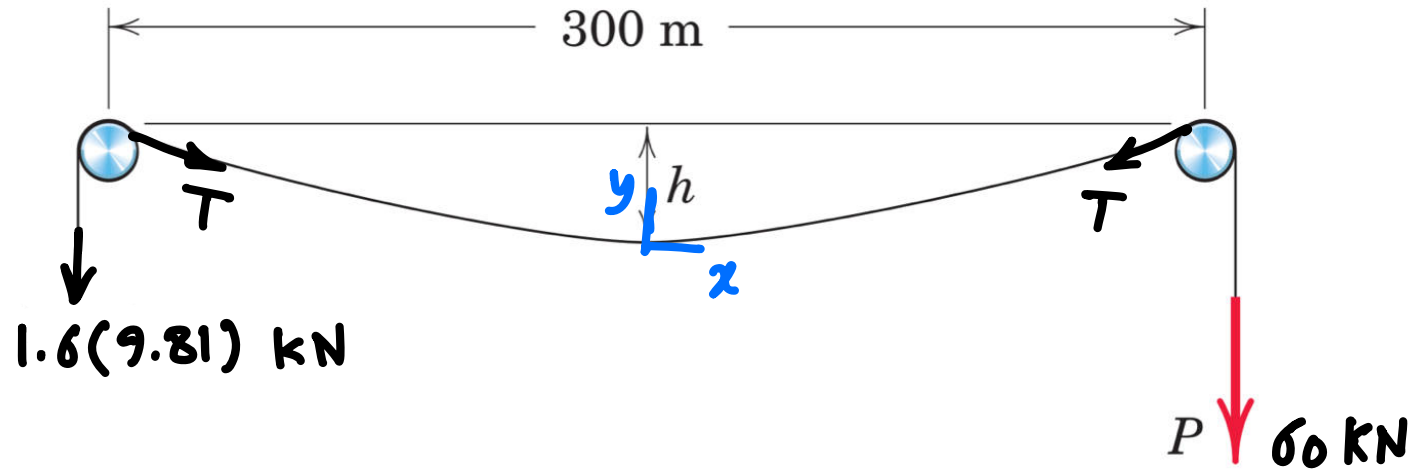
Solution P7.

$$\begin{cases} 15.7 e^{\mu\beta} = T \\ 60 = T e^{\mu\beta} \end{cases}$$

$$\rightarrow 60 = T \left(\frac{T}{15.7} \right) \rightarrow T = 30.7 \text{ kN}$$

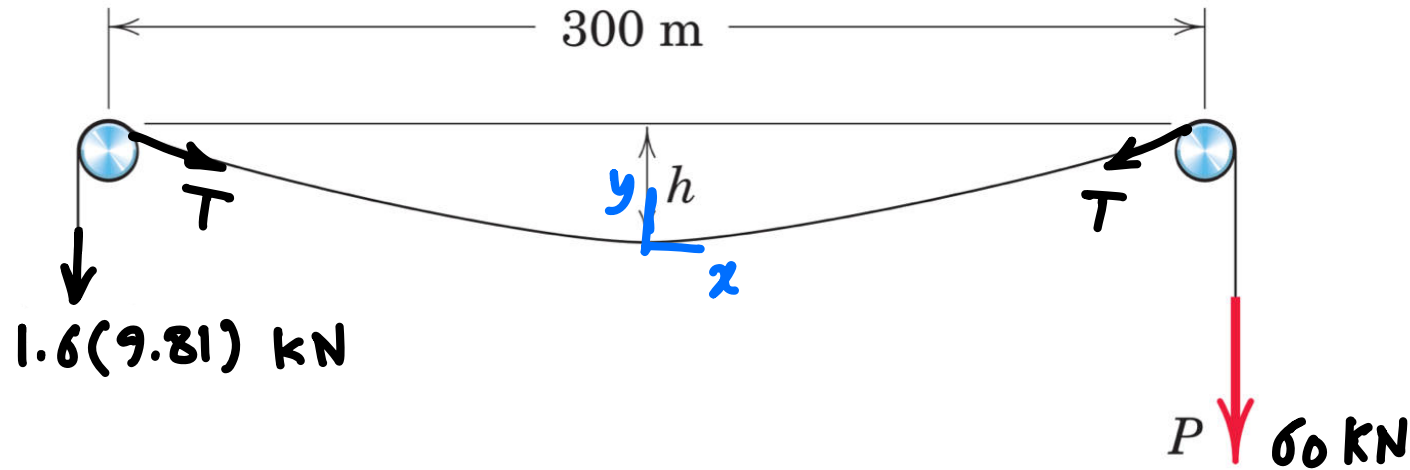
$$15.7 e^{\mu\beta} = 30.7 \rightarrow \mu\beta = 0.670$$



Solution P7.

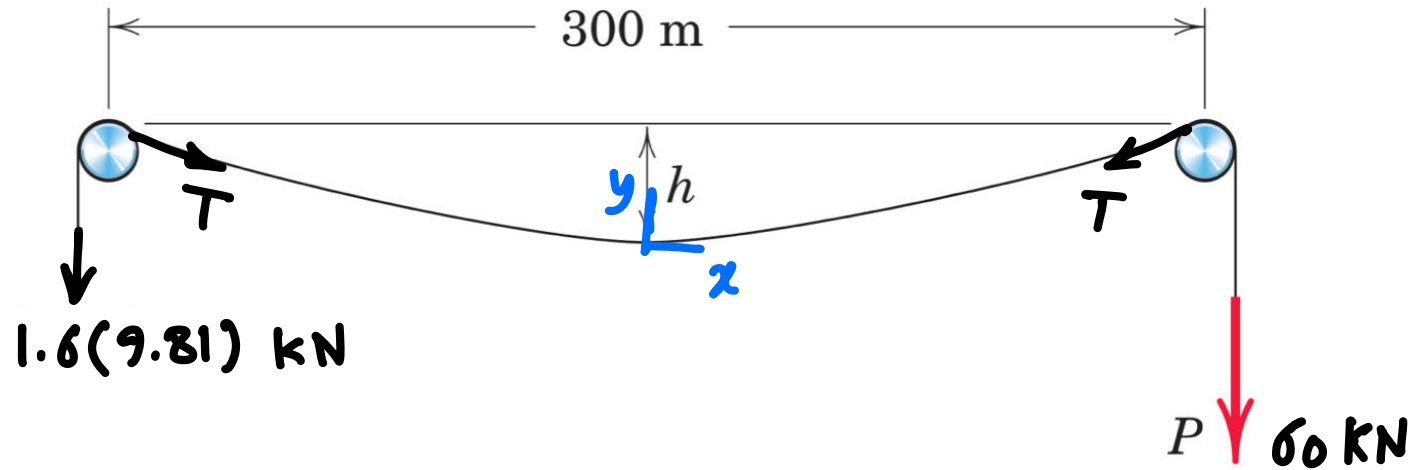
$$T = T_0 \cosh\left(\frac{Pgx}{T_0}\right) ; \quad 30.7 = T_0 \cosh\left(\frac{12(9.81)(0.15)}{T_0}\right)$$

$$\rightarrow T_0 = 23.8 \text{ kN}$$

Solution P7.

$$\frac{dy}{dx} = \sinh \frac{\rho g x}{T_0} \rightarrow \tan \theta = \frac{dy}{dx} \rightarrow \theta = 0.681 \text{ rad}$$

$$\beta = 2.25 \text{ rad} \quad \mu_k = \frac{0.67}{2.25} = 0.293 \approx 0.3$$

Solution P7.

To find $h = ?$

$$T_0 + \rho g h = T$$

$$23.8 + 12(9.81)h \times 10^{-3} = 30.7 \rightarrow h = 58.1 \text{ m}$$